

PRACTICE WORKSHOP

Workshop 5: Polymorphism & Interfaces

This workshop practices writing code against abstractions -- interfaces and superclass references -- so that different concrete classes can be swapped in without changing calling code.

Learning Objectives

- Define and implement an interface with multiple implementing classes.
- Store subclass instances in a superclass/interface-typed collection.
- Rely on dynamic method dispatch instead of instanceof chains where possible.
- Compare interfaces and abstract classes and explain when to use each.

Exercises

Exercise 1: Payable Interface

Define a Payable interface with a method `double getPaymentDue()`. Implement it in both an Invoice class (quantity * pricePerItem) and an Employee class (monthly salary). Write a method that accepts a Payable[] array and prints the total of getPaymentDue() across all elements, without knowing their concrete type.

Hints:

- The array element type should be Payable, not Object.
- Loop with a for-each and call getPaymentDue() polymorphically.

Exercise 2: Notification System

Create a Notification interface with a `send(String message)` method. Implement EmailNotification and SmsNotification classes that each print a differently formatted message. Write a NotificationService class that holds a list of Notification objects and calls send() on each of them for a given message.

Hints:

- Use List to store mixed implementations.

Exercise 3: Sorting Strategy

Implement the Comparable interface on a Product class (compareTo by price), then sort an ArrayList using Collections.sort(). Then write a second sort that orders by name using a Comparator instead, without modifying the Product class.

Hints:

- Comparable affects natural ordering; Comparator is for an alternative ordering.

Stretch Goal (optional)

Add a `DiscountedInvoice` class that extends `Invoice` and overrides `getPaymentDue()` to apply a 10% discount, then confirm your `Payable[]` total-calculating method still works unchanged.

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